Marmara University
Electrical and Electronics Engineering Department
Fall 2023
EE 4092 – Optoelectronics
Term Project - Due January 12, 2023

LUX meter Design

The objective of this project is to design a conceptual luminous meter and test it by simulation or in the lab.

Introduction

A luminous (LUX) meter is a device that measures the "brightness" of a light source and outputs the numerical value pertaining to the source luminous. This is basically achieved by measuring the source optical power and converting this data to luminous, taking into account the luminous efficiency versus wavelength. The block diagram in figure 2 summarizes the device function.



Figure 1: A commercially available LUX meter.

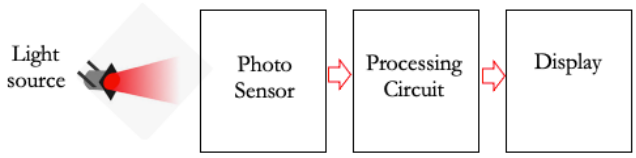


Figure 2: A block diagram of the required device.

The light source is a semi-monochromatic (single color) source with very small spectral width. Enough information about this source should be known (datasheet), including the PI curve, peak wavelength and IV curve. The photo-sensor is to collect the light and make it available to the processing circuit that will, in turn, make the necessary calculations/conversions. The sensitivity of the photosensor and its conversion method need to be known and understood. The processing circuit should take into account the sensor behavior and make the appropriate conversions. The results, in standard luminous units, should be displayed as a decimal number.

Required Work

1. Produce a detailed circuit diagram of your design. Make a list of equipment/parts needed. For each item, the datasheet should be obtained and included.
2. Describe, in details, the working principles of your design. In case programmable units are used, give all details and coding.
3. Discuss the various technical issues involved in the design.
4. Give simulation results for your final product and show the validity of the results.
5. Implement your design and give demonstration (optional step with bonus marks).

Instructions

- Students in the assigned teams work together and submit one report.
- Submissions should be typed and in a formal report format.
- Prepare your project report (one file) in pdf format.
- In case of copied work (analysis, graphs, etc., even partially), all involved parties will get nulls.

Upload link: https://www.dropbox.com/request/MBOYCsa5p2VonLcgi6uh

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Show all work and analysis
Only pdf files are acceptable
Late submissions are checked but not graded
Copied submissions represent academic dishonesty and are disqualified